

Linux Practical questions — Question Paper 3

1. Display the current working directory with its full path

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week$ ls
'Practical Assignment01 linux exercises.docx'  sales_data.txt
'Practical Assignment01 R session Tasks.docx'  sample_data.csv
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week$ pwd
/home/ibab/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week$
```

2. Make a new directory named logs in your home directory

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week$ mkdir logs
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week$ ls
logs
'Practical Assignment01 linux exercises.docx'  'Practical Assignment01 R session Tasks.docx'  sample_data.csv
sales_data.txt
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week$
```

3. Create a file called app.log inside logs/

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week$ cd logs/
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ ls
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ touch app.log
app.log
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$
```

4. Copy app.log to app_backup.log in the same folder

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ cp app.log app_backup.log
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ ls
app_backup.log  app.log
```

5. Move app_backup.log to your Documents folder

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ mv ~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs/app_backup.log ~/Documents/
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ cd
(base) ibab@IBAB-MSc-BDB-Comp01:~$ cd Doc
Docking_Project/ Documents/
(base) ibab@IBAB-MSc-BDB-Comp01:~$ cd Documents/
(base) ibab@IBAB-MSc-BDB-Comp01:~/Documents$ ls
Algorithms          Lab10          ML_lab
AnyDesk             Lab11          ML_Lab_05
app_backup.log      Lab12          'My Practical'
app.log            Lab13          Prep_DB
Bonafide_certificate Lab14          Statement_of_purpose.pdf
'Dataset (1)'       'Lab 3 and Lab 4 - Keshav.tar.xz' studysync
DB-Lab5            Lab8           'Transcript compressed (1).pdf'
'From Scratch'     Lab9           untitled_0_1.odt
git_tests          Letter%20of%20Academic%20Recommendation.odt_0.odt untitled_0.odt
html              'Linux Quiz.pdf' untitled_1.odt
ialsql            MEGA_12
Keshav Pande Resume.pdf ML
(base) ibab@IBAB-MSc-BDB-Comp01:~/Documents$
```

6. Delete app_backup.log from the Documents folder

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/Documents$ rm app_backup.log
(base) ibab@IBAB-MSc-BDB-Comp01:~/Documents$ ls
Algorithms          Lab10                ML
AnyDesk             Lab11                ML_lab
app.log             Lab12                ML_Lab_05
Bonafide_certificate Lab13                'My Practical'
'Dataset (1)'       Lab14                Prep_DB
DB-Lab5            'Lab 3 and Lab 4 - Keshav.tar.xz' Statement_of_purpose.pdf
'From Scratch'     Lab8                studysync
git tests          Lab9                'Transcript compressed (1).pdf'
html               Letter%20of%20Academic%20Recommendation.odt_0.odt  untitled_0_1.odt
ialsql             'Linux Quiz.pdf'    untitled_0.odt
Keshav_Pande_Resume.pdf MEGA_12             untitled_1.odt
(base) ibab@IBAB-MSc-BDB-Comp01:~/Documents$
```

7. List all files in logs/, including hidden files

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ ls -al
total 8
drwxrwxr-x 2 ibab ibab 4096 Jul 20 22:18 .
drwxrwxr-x 3 ibab ibab 4096 Jul 20 22:11 ..
-rw-rw-r-- 1 ibab ibab  0 Jul 20 22:11 app.log
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$
```

8. Show the first 3 lines of app.log

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ head -n 3 app.log
2026-07-20 10:15:05 [INFO] (AuthService) User 'johndoe_12' successfully logged in from IP: 192.168.1.45
2026-07-20 10:16:12 [DEBUG] (CartController) Item 'prod_994' added to cart. Current total: $45.00
2026-07-20 10:18:30 [WARN] (PaymentGateway) API request to Stripe delayed, retrying connection...
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$
```

9. Find all lines containing the word “warning” in a file named server.log

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ grep -i "WARN" app.log
2026-07-20 10:18:30 [WARN] (PaymentGateway) API request to Stripe delayed, retrying connection...
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$
```

10. Replace “debug” with “info” in app.log using sed

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ sed "s/DEBUG/INFO/g" app.log
2026-07-20 10:15:05 [INFO] (AuthService) User 'johndoe_12' successfully logged in from IP: 192.168.1.45
2026-07-20 10:16:12 [INFO] (CartController) Item 'prod_994' added to cart. Current total: $45.00
2026-07-20 10:18:30 [WARN] (PaymentGateway) API request to Stripe delayed, retrying connection...
2026-07-20 10:19:15 [ERROR] (DatabaseService) ConnectionTimeoutException: Failed to connect to MySQL database at 10.0.0.5
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$
```

11. Change permissions of app.log to allow the owner full access

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ chmod u+x app.log
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ ls -l
total 4
-rwxrwxr-- 1 ibab ibab 422 Jul 20 22:22 app.log
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$
```

12. View the permissions and ownership of files in the logs/ directory

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ chmod u+x app.log
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ ls -l
total 4
-rwxrwxr-- 1 ibab ibab 422 Jul 20 22:22 app.log
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$
```

13. Search your home directory for a file named index.html

```
(base) ibab@IBAB-MSc-BDB-Comp01:~$ find . -iname "index.html"
./scholar-notes/frontend/index.html
./R/x86_64-pc-linux-gnu-library/4.5/isoband/doc/index.html
./R/x86_64-pc-linux-gnu-library/4.5/tidyr/doc/index.html
./R/x86_64-pc-linux-gnu-library/4.5/yaml/doc/index.html
./R/x86_64-pc-linux-gnu-library/4.5/ggplot2/doc/index.html
./R/x86_64-pc-linux-gnu-library/4.5/cpp11/doc/index.html
./R/x86_64-pc-linux-gnu-library/4.5/dplyr/doc/index.html
./R/x86_64-pc-linux-gnu-library/4.5/data.table/doc/index.html
./R/x86_64-pc-linux-gnu-library/4.5/openssl/doc/index.html
./R/x86_64-pc-linux-gnu-library/4.5/httptr/doc/index.html
./R/x86_64-pc-linux-gnu-library/4.5/readxl/doc/index.html
./R/x86_64-pc-linux-gnu-library/4.5/tidyselct/doc/index.html
./R/x86_64-pc-linux-gnu-library/4.5/utf8/doc/index.html
```

14. Count how many lines are in app.log

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ wc -l app.log
4 app.log
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$
```

15. Show unique lines in app.log

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ sort -u app.log
2026-07-20 10:15:05 [INFO] (AuthService) User 'johndoe_12' successfully logged in from IP: 192.168.1.45
2026-07-20 10:16:12 [DEBUG] (CartController) Item 'prod_994' added to cart. Current total: $45.00
2026-07-20 10:18:30 [WARN] (PaymentGateway) API request to Stripe delayed, retrying connection...
2026-07-20 10:19:15 [ERROR] (DatabaseService) ConnectionTimeoutException: Failed to connect to MySQL database at 10.0.0.5
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$
```

16. Display all processes currently running on your system

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ ps aux
```

USER	PID	%CPU	%MEM	VSZ	RSS	TTY	STAT	START	TIME	COMMAND
root	1	0.0	0.0	168528	10232	?	Ss	Jun11	1:43	/sbin/init splash
root	2	0.0	0.0	0	0	?	S	Jun11	0:01	[kthreadd]
root	3	0.0	0.0	0	0	?	S	Jun11	0:00	[pool_workqueue_release]
root	4	0.0	0.0	0	0	?	I<	Jun11	0:00	[kworker/R-rcu_g]
root	5	0.0	0.0	0	0	?	I<	Jun11	0:00	[kworker/R-rcu_p]
root	6	0.0	0.0	0	0	?	I<	Jun11	0:00	[kworker/R-slub_]
root	7	0.0	0.0	0	0	?	I<	Jun11	0:00	[kworker/R-netns]
root	9	0.0	0.0	0	0	?	I<	Jun11	0:00	[kworker/0:0H-events_highpri]
root	12	0.0	0.0	0	0	?	I<	Jun11	0:00	[kworker/R-mm_pe]
root	13	0.0	0.0	0	0	?	I	Jun11	0:00	[rcu_tasks_kthread]
root	14	0.0	0.0	0	0	?	I	Jun11	0:00	[rcu_tasks_rude_kthread]
root	15	0.0	0.0	0	0	?	I	Jun11	0:00	[rcu_tasks_trace_kthread]
root	16	2.1	0.0	0	0	?	S	Jun11	1190:12	[ksoftirqd/0]
root	17	0.0	0.0	0	0	?	I	Jun11	16:15	[rcu_preempt]
root	18	0.0	0.0	0	0	?	S	Jun11	0:16	[migration/0]
root	19	0.0	0.0	0	0	?	S	Jun11	0:00	[idle_inject/0]
root	20	0.0	0.0	0	0	?	S	Jun11	0:00	[cpuhp/0]
root	21	0.0	0.0	0	0	?	S	Jun11	0:00	[cpuhp/2]
root	22	0.0	0.0	0	0	?	S	Jun11	0:00	[idle_inject/2]
root	23	0.0	0.0	0	0	?	S	Jun11	0:17	[migration/2]
root	24	0.0	0.0	0	0	?	S	Jun11	0:09	[ksoftirqd/2]
root	26	0.0	0.0	0	0	?	I<	Jun11	0:00	[kworker/2:0H-events_highpri]
root	27	0.0	0.0	0	0	?	S	Jun11	0:00	[cpuhp/4]

17. Find the PID of the cron process

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ ps aux | grep cron
root      731   0.0   0.0   9496  1004 ?        Ss   Jun11   0:09 /usr/sbin/cron -f -P
ibab     190603   0.0   0.0   9216  2652 pts/0    S+   22:40   0:00 grep --color=auto cron
```

18. Kill a process with a specific PID

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ sudo kill -9 731
[sudo] password for ibab:
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ ps aux | grep cron
root      190618   0.0   0.0   9496  3092 ?        Ss   22:41   0:00 /usr/sbin/cron -f -P
ibab     190622   0.0   0.0   9084  2680 pts/0    S+   22:41   0:00 grep --color=auto cron
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ |
```

19. Create a tar archive named logs_archive.tar containing logs/

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ tar -cvf logs_archive.tar logs/
tar: logs: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ LS
LS: command not found
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ la
app.log logs_archive.tar
```

20. Extract the contents of logs_archive.tar to logs_restore/

```
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ ls
app.log logs_archive.tar
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ mkdir -p logs_restore
tar -xvf logs_archive.tar -C logs_restore/
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ ls
app.log logs_archive.tar logs_restore
(base) ibab@IBAB-MSc-BDB-Comp01:~/BBDP307/Assignment/Assignment-1/BDB_NGS_Assignments_first_week/logs$ |
```